900.5.1 DASHBOARD SPECIFICATION v1.0 – VISUAL INTERFACE

900.5.1 EFU DASHBOARD SPECIFICATION – COMPLETE VISUAL INTERFACE

Classification: Noocratic Governance – Operational Interface
Main Function: Real-time Civilizational Diagnostics Visualization

Version: 1.0 | 2026-03-06
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Related Documents:
900.5 (Hungary Pilot)
200.5 (Software Specification)
104.60 (CEWS)
900.1 (CDS)

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1. PURPOSE AND SCOPE

The **900.5.1 Dashboard** is more than a set of charts; it functions as an operational nervous-system interface through which **Fire Chiefs, researchers, and decision-makers** can:

1. Monitor the health of civilization in real time (CII, VKI, SII, SS_total)
2. Detect early warning signals (CEWS acceleration indicators)
3. Analyze trends (historical data and projections)
4. Simulate scenarios (e.g., “What would happen if Nigeria’s TFR declined to 3.0?”)
5. Initiate interventions (activation of 700.x regenerative protocols)

Key Question:
How can complex biophysical data (104.x modules, 600.x Antiflux, and 700.x Regenerative systems) be presented in a visually comprehensible and operationally actionable format?

Answer:
Through a hierarchical, progressive interface that allows navigation from high-level overview to deep module analysis.

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2. DESIGN PHILOSOPHY

2.1 Progressive Disclosure

LEVEL 1 (OVERVIEW)
Dashboard home → Entity health overview (CII, VKI, SII color-coded)

LEVEL 2 (CATEGORY)
Click category → Planetary modules (104.44–46), Demography (104.39), etc.

LEVEL 3 (MODULE)
Click module → Detailed data
Example: 104.52 PFAS – P-LOD values, CFI-B chart, audit matrix

LEVEL 4 (RAW DATA)
Export button → CSV / JSON download for researchers

Principle:
"Display the minimum necessary information while enabling access to maximum analytical depth."

2.2 Color-Coded Alert System

 GREEN – Safe and stable (no intervention required)
 YELLOW – Warning; enhanced monitoring recommended
 RED – Critical condition requiring immediate intervention (Fire Chief protocol activation)
 BLACK – Irreversible collapse zone (emergency protocols only)

Applied to:

• CII, VKI, SII values
• SS_total (planetary balance indicator)
• W_irrev (irreversibility weight)
• CEWS zones (104.60)
• Module outputs (104.x, 600.x, 700.x)

2.3 Real-Time vs. Historical Toggle

REAL-TIME MODE

• Current system values (last update: 2024-03-06 12:35 UTC)
• Live alerts (e.g., CEWS Yellow Zone triggered by VKI acceleration)
• WebSocket updates from pilot-region sensors

HISTORICAL MODE

• Time-series charts (2015–2024)
• Trend analysis (visualization of dCII/dt, dVKI/dt)
• Projection overlays (2030 and 2050 scenarios)

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3. DASHBOARD LAYOUT – WIREFRAME

3.1 Header (Global Navigation)

EFU DASHBOARD | [Search Entity] | □ User

Home | Entities | Modules | Scenarios | Alerts

3.2 Home Screen (Entity Overview)

ENTITY: Hungary (HU-2024) | [Change ▼]

Last update: 2024-03-06 12:35 UTC

CIVILIZATIONAL HEALTH OVERVIEW

CII (Integrity): 0.76 □
VKI (Coherence): 0.68 □ □□ Declining
SII (Instability): 0.42 □
SS_total (Balance): −8.5 □ Moderate deficit

CEWS STATUS:
□ YELLOW ZONE | Alert: VKI acceleration negative ($d^2VKI/dt^2 < 0$)

QUICK METRICS

Population: 9.7M
Area: 93,030 km²

EFU-H (Energy): 60M/year
W_irrev: 420

EFU-M (Water): 45M/year
GDP (reference): $180B

MODULE STATUS (click to expand)

 104.39 Biosphere (Demography) – Stable
 104.44 Pedosphere (Soil) – SS: −8.2
 104.45 Hydrosphere (Water) – SS: −6.5
 104.46 Atmosphere (Climate) – SS: −9.0
 600.52 PFAS Contamination – P-LOD: 3.2

Options:

View Detailed Report | Run Scenario | Export Data

AUTOMATION PANEL

Robot Density
Human Displacement
Automation Productivity
Metabolic Unemployment
ASH Stability Indicator

3.3 CEWS Widget (104.60 Integration)

CEWS – Civilization Early Warning System

Region: Hungary (HU-2024)

Current Status:
 Yellow Zone (Warning)

Metric Trends (2020–2024)

CII: 0.76
Velocity: −0.01/year
Acceleration: +0.001/year²

VKI: 0.68
Velocity: −0.03/year
Acceleration: −0.005/year²

SII: 0.42
Velocity: +0.08/year
Acceleration: +0.02/year²

Active Alerts

□□ VKI Acceleration Critical
($d^2VKI/dt^2 = -0.005/year^2$)
→ Monitor political polarization

□□ 600.52 PFAS P-LOD Elevated (3.2 ng/L)
→ Source audit and filtration upgrade recommended

Options:

View Historical Trends | Run Future Scenario

3.4 Module Deep-Dive (Example: 600.52 PFAS)

600.52 PFAS & Persistent Chemical Audit

Entity: Hungary (HU-2024) – Budapest Pilot

PFAS Load Density (P-LOD)

Drinking Water: 3.2 ng/L □
Blood (average): 2.8 ng/mL □
Urban Soil: 8.5 ng/g □

Chemical Fragmentation Index (CFI-B)

CFI-B = (P-LOD × B-ACC × I-BLOCK) / R × W_irrev

Current CFI-B: 4.2 □

Components:

P-LOD: 3.2 ng/L
B-ACC: 1.5
I-BLOCK: 25%
R_future: High
W_irrev: 950 / 1000

Fire Chief Protocol Recommendations

Immediate actions:

Source audit
Filtration upgrade

Short-term (1–3 months):

Public communication strategy
Procurement ban on PFAS-containing products

Long-term (1 year):

Target P-LOD < 1.0 ng/L
Quarterly monitoring

Options:

View Audit Matrix | Export Report | Activate Protocol

3.5 Scenario Simulator Interface

SCENARIO SIMULATOR
Entity: Nigeria (NG-2024)

Baseline (2024)

Population: 27M
TFR: 6.7

SS_soil: −10.5 □
SS_water: −8.2 □

Scenario Parameters

TFR (2030): 4.5
Renewable energy share: 30%
Education investment: +50%

Projected Results (2050)

Population: 48M (vs. 66M baseline)

SS_soil: −12.0 □ (vs. −18.9 baseline)
SS_water: −9.5 □ (vs. −15.2 baseline)

CII: 0.65 □ (vs. 0.52 baseline)

Interpretation

Lower fertility combined with increased educational investment reduces demographic pressure and stabilizes ecological deficits.

Recommendation:
Invest in **300.3 Educational Flux** and **700.5 Participatory Budgeting**.

Options:

Save Scenario | Export Report | Share Link

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4. TECHNICAL IMPLEMENTATION

4.1 Frontend Stack

Framework: React.js / Next.js

UI Library:
Tailwind CSS
shadcn/ui
Radix UI

Visualization:

Recharts
D3.js
Chart.js

State Management:

Zustand
React Query

Real-time communication:

WebSocket
Server-Sent Events (SSE)

4.2 Component Architecture

Example: **CEWS Widget (React)**

```
import { useQuery } from '@tanstack/react-query';
import { LineChart, Line, XAxis, YAxis, Tooltip } from 'recharts';
```

(Further implementation details remain unchanged and describe component logic.)